

Hyrje në Algoritme

Stacks and Queues - Detyra të zgjidhura & detyra për ushtrime

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Asistentët:

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Rreze Rexhepi

Nëntor, 2024

Prishtinë

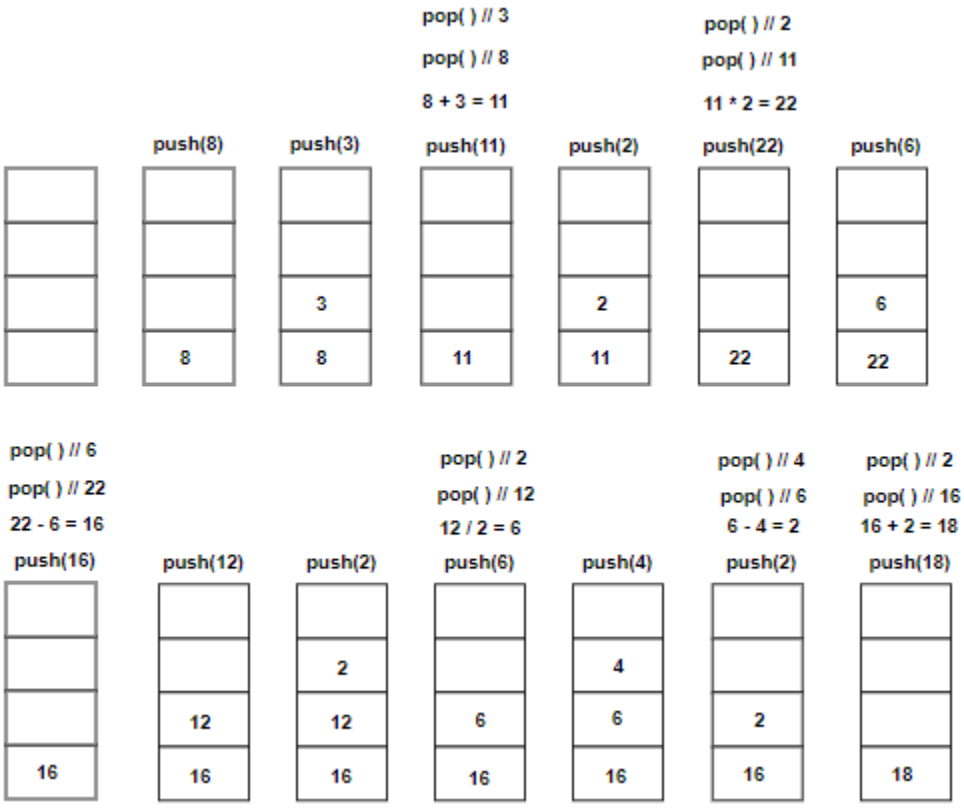
Detyra të zgjidhura

Detyra 1. Të zgjidhet detyra hap pas hapi duke përdorur One-Stack.

((8 + 3) * 2 - 6) + (12 / 2) - 4

Zgjidhje:

Postfix: 8 3+2*6-12 2/ 4-+

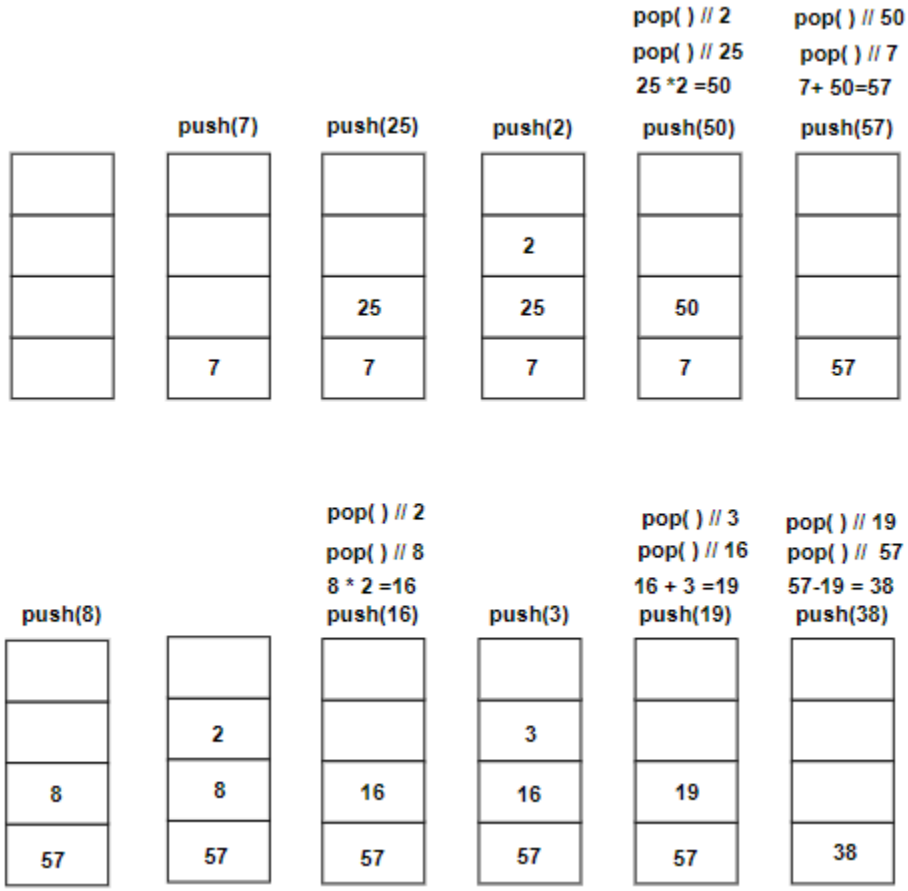


Detyra 2. Të zgjidhet detyra hap pas hapi duke përdorur One-Stack.

7 + (25 * 2) - ((8 * 2) + 3)

Zgjidhje:

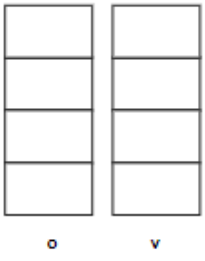
Postfix: 7 25 2*+8 2*3+-



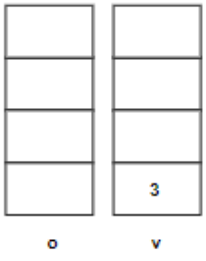
Detyra 3: Të kryhen operacionet në vijim duke përdorur Dijkstra 2-Stack:

((3 + (6 * 4) - ((8 * 2 / 4)))

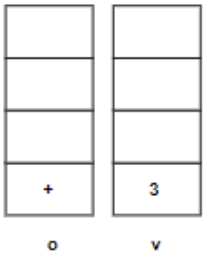
1)



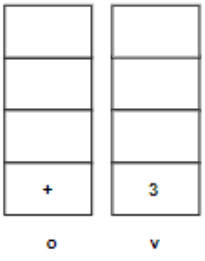
2) v.push(3)



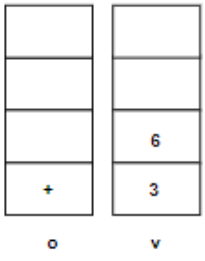
3) o.push(+)



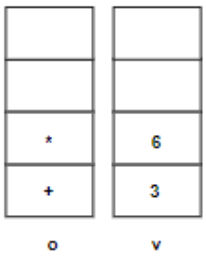
4) (kllapat hapëse injorohen)



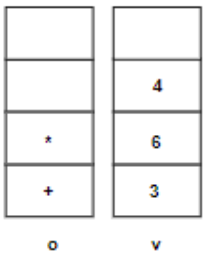
5) v.push(6)



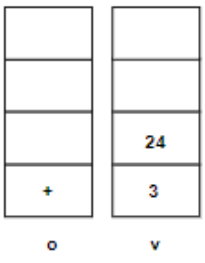
6) o.push(*)



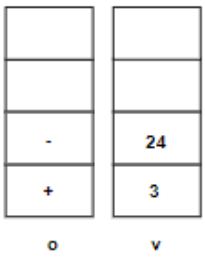
7) v.push(4)



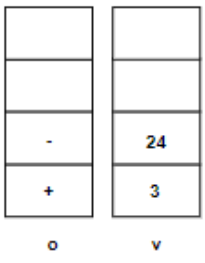
8) v.pop( ) // 4
v.pop( ) // 6
o.pop( ) // *
6 * 4 = 24
v.push(24)



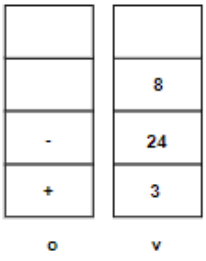
9) o.push(-)



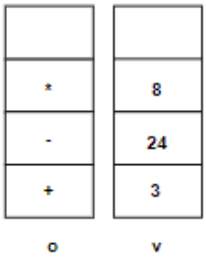
10) (kllapat hapëse injorohen)



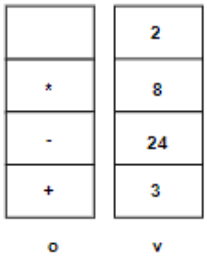
11) v.push(8)



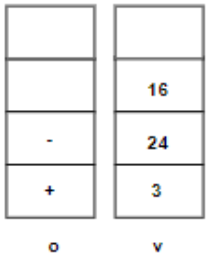
12) o.push(*)



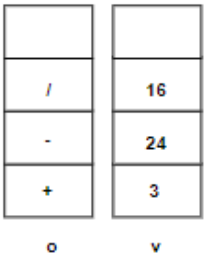
13) v.push(2)



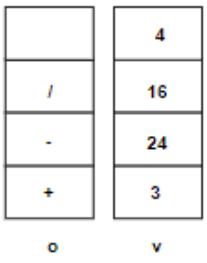
14) v.pop( ) // 2
v.pop( ) // 8
o.pop( ) // *
8 * 2 = 16
v.push(16)



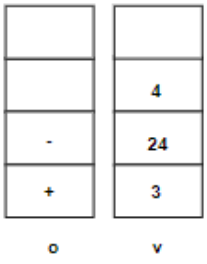
15) o.push(/)



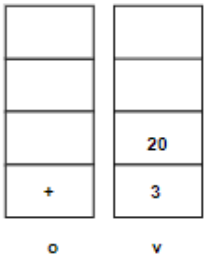
16) v.push(4)



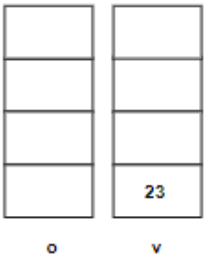
17) v.pop( ) // 4
v.pop( ) // 16
o.pop( ) // /
16 / 4 = 4
v.push(4)



18) v.pop( ) // 4
v.pop( ) // 24
o.pop( ) // -
24 - 4 = 20
v.push(20)



19) v.pop( ) // 20
v.pop( ) // 3
o.pop( ) // +
3 + 20 = 23
v.push(23)

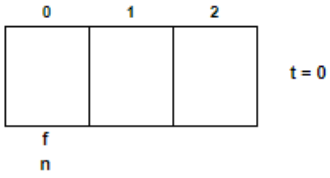


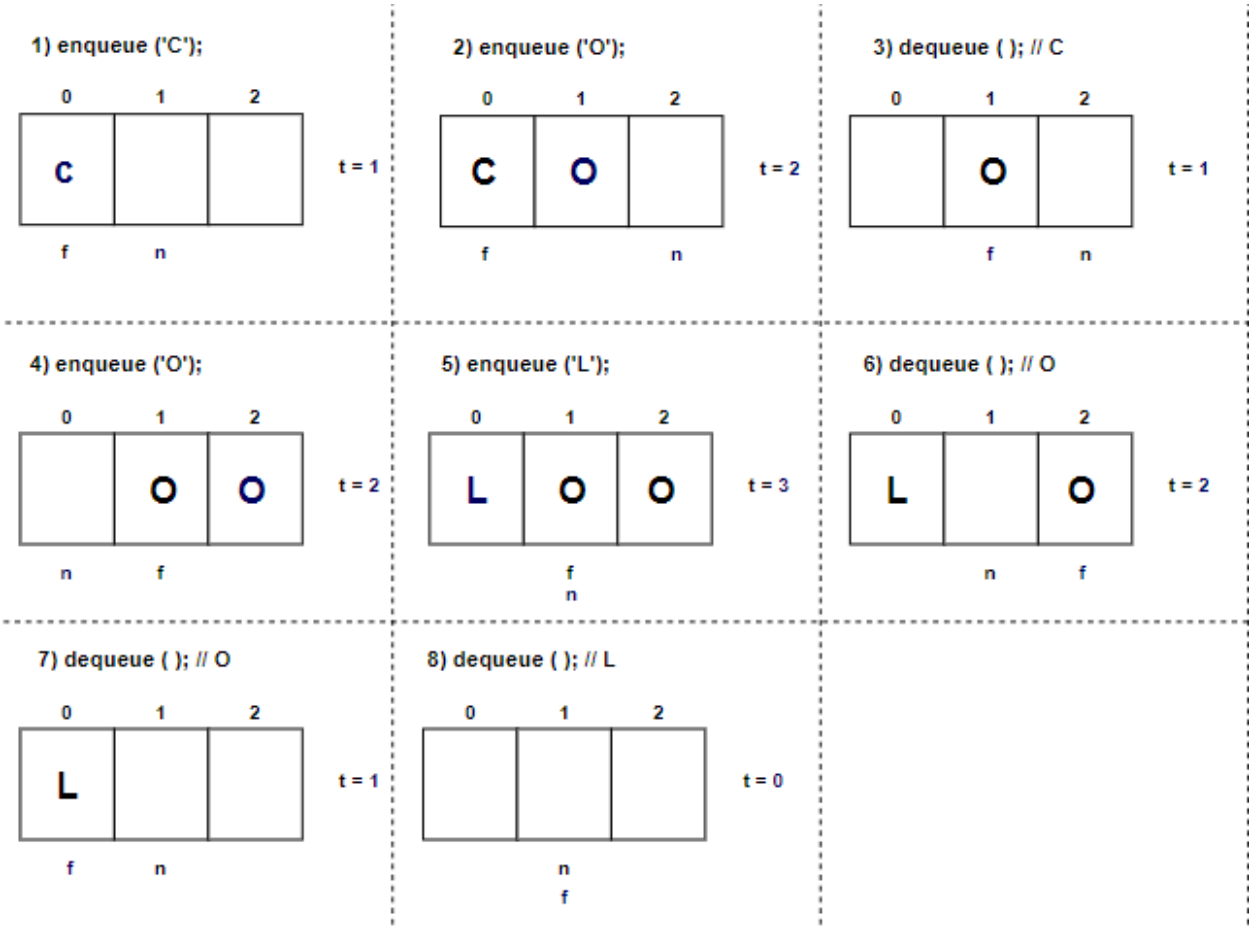
Detyra 4. Të zgjidhet detyra duke përdorur queue të madhësisë **3**. Të paraqiten grafikisht të gjithë hapat.

Fjala **COOL**

Zgjidhje:

Vlera fillestare:





Detyra për ushtrime

Detyra 1. Të kthehen shprehjet e mëposhtme nga infix në postfix:

1. A + B / (C * D) – E
2. 5 * 6 – 8 + 9 * 7
3. (A * B) / (C – D * E) + F
4. a / b – c / d + e * f / g * h + i

Detyra 2. Të zgjidhet detyra hap pas hapi duke përdorur One-Stack. Numrat X (integer) të zëvendësohen sipas 5 shifrave të fundit të ID-së suaj.

1. X * X – X + X / X (p.sh. nëse ID = 222312345 -> 1 * 2 – 3 + 4 / 5)
2. X - X + (X / X) * X (p.sh. nëse ID = 222312345 -> 1 – 2 + 3 / 4 * 5)

Detyra 3. Të zgjidhet detyra hap pas hapi duke përdorur Dijkstra Two-Stack.

1. ((3 * 2) / (1 + 2))
2. ((2 + (4 * 5)) – ((8 * 2) / 4))

Detyra 4. Të realizohen operacionet në një queue të madhësisë **3** sipas renditjes së mëposhtme. Të paraqiten grafikisht të gjithë hapat.

Enqueue(**I**);

Enqueue(**N**);

Dequeue();

Enqueue(**P**);

Enqueue(**U**);

Dequeue();

Dequeue();

Enqueue(**T**);

Dequeue();

Detyra 5. Të zgjidhet detyra duke përdorur queue të madhësisë **4**. Të paraqiten grafikisht të gjithë hapat.

Emri dhe mbiemri juaj (p.sh. **FILANFISTEKU**)

Detyra 6. Të zgjidhet detyra duke përdorur queue të madhësisë **2**. Të paraqiten grafikisht të gjithë hapat.

Vendlindja juaj (p.sh. **GJILAN**)